

CHAPTER 2

Part I – Foundations of Identity-Centric Security

Chapter 2

Federal Identity, Credential, and Access Management (FICAM): Identity Governance Across Federal, Defense, and Mission Environments

1. Chapter Purpose

Chapter 2 establishes the governance, policy, and architectural framework that underpins identity security across the United States Federal Government. Readers examine the evolution of Federal Identity, Credential, and Access Management (FICAM), its relationship to Identity, Credential, and Access Management (ICAM), and its integration with Zero Trust Architecture. By understanding the governance principles that drive identity modernization, readers gain the policy and operational context necessary to appreciate how Active Directory supports secure authentication, authorization, federation, and identity lifecycle management within federal agencies, Defense Agencies, Combatant Commands, and mission partner environments.

2. Chapter Abstract

Identity security is not solely a technical discipline; it is a governance discipline. Before enterprise identity infrastructures can be designed, secured, monitored, or defended, they must operate within a framework that defines how digital identities are established, trusted, authenticated, authorized, and managed throughout their lifecycle. Within the United States Federal Government, this framework is provided through Federal Identity, Credential, and Access Management (FICAM), which establishes common standards and practices that promote interoperability, security, and trust across diverse mission environments. This chapter examines the historical development of FICAM, from Homeland Security Presidential Directive 12 (HSPD-12) through contemporary Zero Trust initiatives and federal identity modernization efforts. Readers explore identity proofing, credential issuance, federation, privileged access governance, identity lifecycle management, and enterprise trust relationships while examining how these concepts support mission assurance across civilian agencies, defense organizations, and interagency operations.

The chapter concludes by demonstrating how FICAM principles are implemented through enterprise identity technologies such as Active Directory Domain Services, Active Directory Certificate Services, Microsoft Entra ID, and federation services. These technologies form the operational identity infrastructure examined throughout the remainder of the book.

3. Key Terminology

- Federal Identity, Credential, and Access Management (FICAM)
- Identity, Credential, and Access Management (ICAM)
- Identity Governance
- Identity Lifecycle Management
- Homeland Security Presidential Directive 12 (HSPD-12)
- Federal Information Processing Standard (FIPS) 201
- Personal Identity Verification (PIV)
- Derived PIV Credential

- Identity Proofing
- Identity Assurance Level (IAL)
- Authenticator Assurance Level (AAL)
- Federation Assurance Level (FAL)
- Federation
- Identity Provider (IdP)
- Relying Party (RP)
- Attribute-Based Access Control (ABAC)
- Role-Based Access Control (RBAC)
- Privileged Access Management (PAM)
- Continuous Diagnostics and Mitigation (CDM)
- Zero Trust Architecture

4. Chapter Learning Objectives

Upon completion, readers will be able to:

- Explain the purpose and objectives of FICAM.
- Differentiate FICAM from enterprise ICAM implementations.
- Describe the federal identity lifecycle from enrollment through deprovisioning.
- Explain identity proofing and credential issuance processes.
- Describe the role of federation within mission partner environments.
- Differentiate RBAC and ABAC authorization models.
- Explain how FICAM supports Zero Trust Architecture.
- Relate identity governance principles to Active Directory implementations.

5. Standards, Frameworks, and Guidance Covered

- FICAM Architecture Framework
- Federal ICAM Architecture
- HSPD-12
- FIPS 201
- NIST SP 800-53
- NIST SP 800-63 (Parts A, B, and C)
- NIST SP 800-207
- NIST Risk Management Framework (RMF)
- NIST Cybersecurity Framework (CSF)
- Executive Order 14028
- OMB M-19-17
- OMB M-22-09
- Federal Zero Trust Strategy
- CISA Zero Trust Maturity Model
- DoD Zero Trust Reference Architecture

6. Included Figures

Figure 2-1. Evolution of Federal Identity Management

Figure 2-2. Components of the FICAM Architecture

Figure 2-3. Federal Identity Lifecycle

Figure 2-4. Credential Issuance and Revocation Process

Figure 2-5. Federation Trust Across Mission Partners

Figure 2-6. Identity Assurance Levels (IAL, AAL, and FAL)

Figure 2-7. Zero Trust Identity Pillar

Figure 2-8. Relationship Between FICAM and Active Directory

Figure 2-9. Enterprise Identity Governance Model

7. Included Tables

Table 2-1. Core Components of FICAM

Table 2-2. Comparison of IAL, AAL, and FAL

Table 2-3. Federal Identity Standards and Their Purpose

Table 2-4. Identity Proofing Methods

Table 2-5. RBAC versus ABAC

Table 2-6. Identity Lifecycle Phases

Table 2-7. FICAM Integration with Active Directory

Table 2-8. Identity Governance Roles Within Federal Mission Environments

8. Included Case Studies

Case Study 2-1. The Evolution of Federal Identity Following HSPD-12

Case Study 2-2. Cross-Agency Federation and Trust Challenges

Case Study 2-3. Identity Modernization Within a Federal Mission Environment

Case Study 2-4. Zero Trust Adoption Across Defense Organizations

9. Included Labs / Exercises

Lab 2-1. Mapping the Federal Identity Lifecycle

Lab 2-2. Building a Sample FICAM Architecture

Lab 2-3. Evaluating Identity Assurance Levels

Lab 2-4. Designing a Federation Trust Model

Lab 2-5. Mapping RBAC and ABAC Permissions

10. Offensive Concepts Covered

- Identity Trust Abuse
- Federation Trust Exploitation (Introduction)
- Credential Targeting
- Authentication Weaknesses
- Authorization Weaknesses
- Insider Threat Considerations
- Identity Attack Surface Analysis

11. Defensive Concepts Covered

- Identity Governance
- Credential Lifecycle Management
- Identity Assurance
- Continuous Authentication
- Zero Trust Identity Principles
- Least Privilege
- Separation of Duties
- Privileged Access Governance
- Identity Risk Management

12. Offensive Tools Discussed

- BloodHound (Identity Relationship Analysis)
- PowerView (Directory Enumeration Preview)
- Certipy (Certificate Services Preview)

- PingCastle (Identity Risk Assessment)

13. Defensive Tools Discussed

- Microsoft Entra ID
- Microsoft Defender for Identity
- Active Directory Certificate Services (AD CS)
- Microsoft Entra Connect
- Active Directory Users and Computers (ADUC)
- PingCastle
- Purple Knight
- Microsoft Identity Manager (Historical Context)

14. Primary FICAM Components Covered

- Identity Management
- Credential Management
- Access Management
- Federation Management
- Identity Proofing
- Authentication Services
- Authorization Services
- Identity Governance
- Lifecycle Management
- Continuous Evaluation

15. MITRE ATT&CK Framework Coverage

Although governance-focused, this chapter introduces identity-related adversary behaviors that exploit weaknesses in authentication, authorization, and trust relationships.

Primary ATT&CK Tactics

- TA0001 – Initial Access
- TA0003 – Persistence
- TA0005 – Defense Evasion
- TA0006 – Credential Access

Representative Techniques

- T1078 – Valid Accounts
- T1550 – Use Alternate Authentication Material
- T1556 – Modify Authentication Process
- T1484 – Domain or Tenant Policy Modification (Conceptual Introduction)

16. Chapter Sections

2.1 The Evolution of Federal Identity Management

2.2 Understanding FICAM and Federal ICAM

2.3 HSPD-12, FIPS 201, and the Modern Federal Identity Landscape

2.4 Identity Proofing, Credential Issuance, and Lifecycle Management

2.5 Authentication, Authorization, Federation, and Trust Relationships

2.6 Identity Governance Models: RBAC, ABAC, and Privileged Access

2.7 Zero Trust Architecture and Identity-Centric Security

2.8 Implementing FICAM Through Active Directory and Enterprise Identity Infrastructure

2.9 FICAM Across Civilian Agencies, Defense Agencies, Combatant Commands, and

Mission Partners

2.10 Chapter Summary

- 2.10.1 Key Concepts Reviewed
- 2.10.2 Key Takeaways
- 2.10.3 Suggested Reading
- 2.10.4 Preparing for Chapter 3